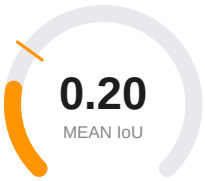
What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

No vessel fits the slick well enough to be named.



Forward replay tested 4 candidates. None recreates the observed slick above the fit gate, so the system returns no name rather than the least-bad one; the source may not be in the data.

BEST FIT	FIT GATE	NEXT BEST	CANDIDATES
0.20	**0.3**	**0.03**	**4**
mean IoU of the best replay	needed to name a vessel	mean IoU of the runner-up	replayed by forward replay

EVIDENCE

- The best fit reaches **0.20**, below the **0.3** gate
- The next-best candidate fits at only **0.03**
- None of the **4** replayed candidates recreates the slick well enough
- The nearest vessel, **419486428**, 0.0014° away, is not named either

Candidates, ranked by physical fit

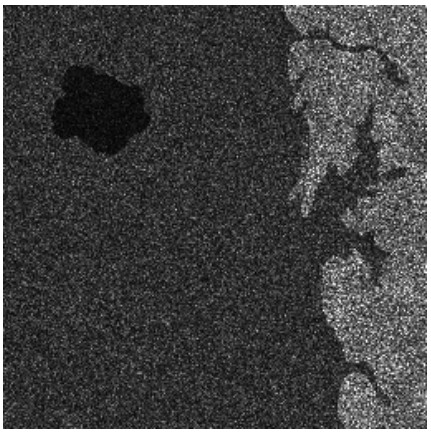
Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

#	VESSEL	MEAN IoU	VARIANTS WON	RELEASE STARTS
1	419486428	0.202	6 of 6	10 Sep 2026, 03:30 UTC
2	419534124	0.031	0 of 6	9 Sep 2026, 16:30 UTC
3	419655571	0.000	0 of 6	9 Sep 2026, 08:30 UTC
4	419833570	0.000	0 of 6	9 Sep 2026, 21:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419486428**, ranked 1 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

100%
LEAD RETAINED

80 tested changes to the model's assumptions; **0** changed the leader.

No tested change moved the leader.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	495 km^2	0%	43%	0%
+12 h 10 Sep 2026, 16:30 UTC	542 km^2	0%	40%	0%
+24 h 11 Sep 2026, 04:30 UTC	590 km^2	0%	28%	0%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

1	Excluded	track data at every queried time, never inside the envelope
4	Candidate	track data present and inside the backward envelope at some queried time

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**79%** 50/63	**82%** 49/60
False alerts	**28%** 27/97	**20%** 16/79
Mistaken promotions	**—** 0/0	**—** 0/0

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:11 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`5db2e4757712bf85f3bfb3e64a0f0d4f1662a04dd7a0e392035c9b866c5a7faf`
coverage_result.json	`2a86fa8542fd2367248ad3990ec57f897ab706e8b51572b29b6839ce3d4636f0`
forcing.npz	`07595f69999ce4c57598ba07aa7a5def993bd877a5709305ca9fc0b3a07b8512`
forecast_result.json	`92efc88a87e99f34101d1ce6c88f4fa9f7bda6fb37a70e9c27a56591a1ebdadb`
fragility_result.json	`4811a6875cef91c32e69f1cdf02405a1141b1c4b587b240314ad1fbca98aa79f`
historical_mapping.json	`f600ca374180a6c5cdb4e1fc9e4316bc3c940b65aa360840827f09a879242a97`
manifest.json	`429be4b4085271a87088cef7e5e628d1f4887e26efc03af8c6368da10818d860`
mask_groundtruth.png	`57f9574e078cb2c6f8ec5732b2771fe34a6f26869fd6ac506187077f4d846387`
naive_baseline_result.json	`ea0683fbf3354ad0cee97e2839333ba4a8e8a7ea209976eadd0b1b6524444861`
next_observation_result.json	`d5654d2e3535eb49d5095c92a56c391ae96927e70600610b823aea1ff2f45cd8`
scene.png	`e81930a0301fbc5e03d9911402bda9656712d08f94052e6cd5cf8e7bc75a9ce8`

Uncertainty

- Rankings express physical consistency under tested assumptions. They are not proof of responsibility.
- Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.
- Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.
- Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.